

# Optimal Taxation with Heterogeneous Agents

Ramsey Model with Workers and Capitalists

Based on Greulich, Laczó & Marcet (2022)

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## 1. Motivation: Adding Heterogeneity

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In the homogeneous-agent Ramsey model, the only question was one of **efficiency**: which tax distorts less? The answer was the Chamley–Judd result:  $\tau_\infty^k = 0$ .

Now we add a second dimension: **distribution**. The central questions become:

- Should tax revenue come from capital or labour taxes?
- What are the efficiency and distribution considerations?
- Can we design a tax reform that makes *everyone* better off (Pareto improving)?

**Intuition:** *With homogeneous agents,  $\tau^k$  high affects everyone equally — the question is only efficiency. With heterogeneous agents,  $\tau^k$  high hurts capitalists more (they have more capital), and  $\tau^\ell$  high hurts workers more (they depend more on their salary). Now the government faces a trade-off between efficiency and distribution that did not exist before.*

## 2. Environment

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We consider a 2-class economy with **workers** and **capitalists**. They differ in their (permanent) labour productivity and initial wealth. Both groups can work and save/borrow.

Each consumer  $j = 1, 2$  solves:

$$\max_{\{c_{j,t}, l_{j,t}\}} \sum_{t=0}^{\infty} \beta^t [u(c_{j,t}) + v(l_{j,t})] \quad (1)$$

subject to the budget constraint:

$$c_{j,t} + k_{j,t} = w_t \phi_j l_{j,t} (1 - \tau_t^\ell) + k_{j,t-1} [1 + (r_t - \delta)(1 - \tau_t^k)], \quad j = 1, 2. \quad (2)$$

**Intuition:** *This is the same structure as the homogeneous model, but now everything carries a subscript  $j$ . The key differences are:*

- $\phi_j$ : type-specific labour productivity. Each group produces a different amount per hour worked. In the homogeneous model,  $\phi = 1$  for everyone.
- $k_{j,-1}$ : type-specific initial wealth. One group starts rich, the other poor (or in debt).

*The taxes  $\tau_t^\ell$  and  $\tau_t^k$  are the same for both groups (flat-rate). The government cannot tax each group differently.*

### What is equal to the homogeneous model?

The preferences have the same structure:  $u(c) + v(l)$ , same  $\beta$  for both groups, same infinite horizon. The firm, the markets, the taxes — all the same. The only novelty is  $\phi_j$  and  $k_{j,-1}$  varying by group.

### What does “workers” and “capitalists” mean?

Both groups can work and save. The labels refer to their *relative* source of income:

- **Capitalists** (group 1): wealthier relative to their labour earnings. More income from capital.
- **Workers** (group 2): higher ratio of labour productivity to initial wealth. More income from labour.

## 3. Firm

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The representative firm produces using capital and effective labour:

$$F(k_{t-1}, e_t)$$

The firm maximizes profits. The optimality conditions are the same as in the homogeneous model:

$$F_k(k_{t-1}, e_t) = r_t \tag{3}$$

$$F_e(k_{t-1}, e_t) = w_t \tag{4}$$

Each factor is paid its marginal product (perfect competition).

## 4. Market Clearing Conditions

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We normalise the mass of each group to 1/2. The market clearing conditions are:

**Labour market:**

$$e_t = \frac{1}{2} \sum_{j=1}^2 \phi_j l_{j,t} \tag{5}$$

**Intuition:** Total effective labour is the average of what each group contributes. The  $1/2$  comes from each group being half the population. Even if both groups work the same hours, their effective labour differs because of different productivities  $\phi_j$ .

**Capital market:**

$$k_t = k_t^g + \frac{1}{2} \sum_{j=1}^2 k_{j,t} \quad (6)$$

**Intuition:** Total capital is the government's capital  $k_t^g$  (which can be negative, meaning debt) plus the average of both groups' savings. Note that  $k_t^g$  is the government's asset position (stock), which is different from  $g$  (the flow of government spending each period).

**Goods market:**

$$\frac{1}{2} \sum_{j=1}^2 c_{j,t} + g + k_t - (1 - \delta)k_{t-1} = F(k_{t-1}, e_t) \quad (7)$$

**Intuition:** Same feasibility constraint as before: production must cover average consumption of both groups, government spending, and investment (including replenishment of depreciated capital).

## 5. Competitive Equilibrium Conditions

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### Euler equation

For each consumer  $j = 1, 2$ :

$$u'(c_{j,t}) = \beta u'(c_{j,t+1}) [1 + (r_{t+1} - \delta)(1 - \tau_{t+1}^k)] \quad (8)$$

### Consumption–labour optimality condition

For each consumer  $j = 1, 2$ :

$$-\frac{v'(l_{j,t})}{u'(c_{j,t})} = w_t \phi_j (1 - \tau_t^\ell) \quad (9)$$

## 6. Functional Forms (Assumption A1)

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$$u(c) = \frac{c^{1-\sigma_c}}{1-\sigma_c} \quad \text{and} \quad v(l) = -\omega \frac{l^{1+\sigma_l}}{1+\sigma_l} \quad (10)$$

where:

- $\omega > 0$ : relative weight of the disutility of hours worked.
- $\sigma_c > 0$ : coefficient of relative risk aversion. Controls how much you dislike consumption variability. Higher  $\sigma_c$  means you strongly prefer smooth consumption over time.

- $\sigma_l > 0$ : inverse of the Frisch elasticity of labour supply. Higher  $\sigma_l$  means your hours barely change when your salary changes (rigid labour supply).

The derivatives are:

$$u'(c) = c^{-\sigma_c} \quad \text{and} \quad v'(l) = -\omega l^{\sigma_l} \quad (11)$$

**Intuition:** *These are the same  $u(c)$  and  $v(l)$  from the homogeneous model, now written with a specific functional form instead of being left generic. They have the same properties:  $u$  is increasing and concave,  $v$  is decreasing. This specific form is standard in macroeconomics because it has convenient mathematical properties (constant elasticities, compatible with balanced growth).*

## 7. Sharing Rules: Consumption and Labour

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Assumption A1 delivers a powerful simplification: **if you know what group 1 does, you automatically know what group 2 does.**

### Consumption sharing rule

$$c_{2,t} = \lambda c_{1,t}, \quad \forall t \quad (12)$$

for some constant  $\lambda$  to be determined in equilibrium.

**Intuition:** *The consumption of group 2 is always a fixed fraction  $\lambda$  of the consumption of group 1, in every period. If  $\lambda = 0.54$ , workers consume 54% of what capitalists consume, and this proportion never changes.*

*This comes from the Euler equation. Both groups face the same prices and taxes. With CRRA utility, their Euler equations imply that the growth rate of consumption is the same for both groups. If both grow at the same rate, the ratio  $c_2/c_1$  never changes. That constant ratio is  $\lambda$ .*

### Labour sharing rule

$$l_{2,t} = \mathcal{K}(\lambda) l_{1,t}, \quad \forall t \quad (13)$$

where

$$\mathcal{K}(\lambda) \equiv \lambda^{-\sigma_c/\sigma_l} \left( \frac{\phi_2}{\phi_1} \right)^{1/\sigma_l} \quad (14)$$

**Intuition:**  $\mathcal{K}(\lambda)$  defines the ratio of labour between group 2 and group 1. It depends only on the primitives  $\sigma_c$ ,  $\sigma_l$ ,  $\phi_1$ ,  $\phi_2$ , and  $\lambda$ .

*If  $\mathcal{K}(\lambda) > 1$ , group 2 works more than group 1. If  $\mathcal{K}(\lambda) < 1$ , group 2 works less. If  $\mathcal{K}(\lambda) = 1$ , both work the same.*

## Derivation of the labour sharing rule

**Step 1:** Write the consumption-labour condition for each group:

$$-\frac{v'(l_{1,t})}{u'(c_{1,t})} = w_t \phi_1 (1 - \tau_t^\ell) \quad -\frac{v'(l_{2,t})}{u'(c_{2,t})} = w_t \phi_2 (1 - \tau_t^\ell)$$

**Step 2:** Divide group 2 by group 1. The wage  $w_t$  and the tax  $\tau_t^\ell$  cancel:

$$\frac{v'(l_{2,t}) \cdot u'(c_{1,t})}{v'(l_{1,t}) \cdot u'(c_{2,t})} = \frac{\phi_2}{\phi_1}$$

**Step 3:** Plug in the derivatives from A1 ( $u'(c) = c^{-\sigma_c}$ ,  $v'(l) = -\omega l^{\sigma_l}$ ). The  $\omega$  cancels:

$$\frac{l_{2,t}^{\sigma_l} \cdot c_{1,t}^{-\sigma_c}}{l_{1,t}^{\sigma_l} \cdot c_{2,t}^{-\sigma_c}} = \frac{\phi_2}{\phi_1}$$

**Step 4:** Substitute  $c_{2,t} = \lambda c_{1,t}$ . The  $c_{1,t}$  terms cancel:

$$\left(\frac{l_{2,t}}{l_{1,t}}\right)^{\sigma_l} \cdot \lambda^{\sigma_c} = \frac{\phi_2}{\phi_1}$$

**Step 5:** Solve for  $l_{2,t}/l_{1,t}$ :

$$l_{2,t} = \underbrace{\lambda^{-\sigma_c/\sigma_l} \left(\frac{\phi_2}{\phi_1}\right)^{1/\sigma_l}}_{\mathcal{K}(\lambda)} \cdot l_{1,t}$$

## Interpreting $\mathcal{K}(\lambda)$ : two competing forces

$\mathcal{K}(\lambda)$  is the product of two pieces, each representing a different force:

**Piece 1:**  $\lambda^{-\sigma_c/\sigma_l}$  — **the poverty effect.**

$\lambda = c_2/c_1$ . If  $\lambda < 1$ , group 2 consumes less (is poorer). The exponent is negative ( $-\sigma_c/\sigma_l$ ), so when  $\lambda$  decreases, this piece increases. Poorer agents work more because each extra euro gives them more happiness (high marginal utility of consumption), so they are more willing to sacrifice leisure for income.

But being poorer does not *necessarily* mean working more — it depends on whether this force is stronger than the next one.

**Piece 2:**  $(\phi_2/\phi_1)^{1/\sigma_l}$  — **the productivity effect.**

If  $\phi_2 < \phi_1$ , group 2 is less productive and earns less per hour. The exponent is positive ( $1/\sigma_l$ ), so when the productivity ratio decreases, this piece decreases. Less productive agents work less because the reward per hour does not compensate the cost of sacrificing leisure.

**The result depends on which force dominates:**

- If the poverty effect dominates:  $\mathcal{K}(\lambda) > 1$ , group 2 works more despite being less productive.

- If the productivity effect dominates:  $\mathcal{K}(\lambda) < 1$ , group 2 works less.
- If they balance:  $\mathcal{K}(\lambda) \approx 1$ , both groups work similarly.

The parameter  $\sigma_l$  controls the sensitivity of both effects. If  $\sigma_l$  is low (elastic labour supply), both effects are amplified. If  $\sigma_l$  is high (rigid labour supply), both effects are dampened.

## 8. Implementability Constraints

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In the homogeneous model there was a single implementability constraint summarising all the optimising behaviour of the private sector. Now with two groups there are **two** — one for each group.

### Implementability constraint for group 1

$$\sum_{t=0}^{\infty} \beta^t [u'(c_{1,t}) c_{1,t} + v'(l_{1,t}) l_{1,t}] = u'(c_{1,0}) k_{1,-1} [1 + (r_0 - \delta)(1 - \tau_0^k)] \quad (7)$$

This is identical in structure to the implementability constraint from the homogeneous model, applied to group 1.

### Implementability constraint for group 2

The implementability constraint for group 2 has the same structure, but originally involves  $c_2$  and  $l_2$ . Since  $c_{2,t} = \lambda c_{1,t}$  and  $l_{2,t} = \mathcal{K}(\lambda) l_{1,t}$ , it is rewritten in terms of group-1 variables:

$$\sum_{t=0}^{\infty} \beta^t \left[ u'(c_{1,t}) \lambda c_{1,t} + \frac{\phi_2}{\phi_1} v'(l_{1,t}) \mathcal{K}(\lambda) l_{1,t} \right] = u'(c_{1,0}) k_{2,-1} [1 + (r_0 - \delta)(1 - \tau_0^k)] \quad (10)$$

**Intuition:** *The path from equation (7) to equation (10) is:*

1. Take the implementability constraint (7), which is written for  $j = 1$ .
2. Write the same structure for  $j = 2$  (with  $c_2$ ,  $l_2$ ,  $k_{2,-1}$ ).
3. Substitute  $c_2 = \lambda c_1$  and  $l_2 = \mathcal{K}(\lambda) l_1$ .
4. What remains is equation (10): the group-2 constraint expressed in group-1 variables.

## 9. Additional Policy Constraints

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In addition to the standard restrictions, the government may face practical or institutional limits on how extreme the taxes and allocations can become. We introduce two extra constraints.

### 9.1. Upper Bound on Capital Taxes

Often there is an exogenous upper limit on how high the capital tax rate can be set, for example due to political economy or tax evasion concerns. Let  $\bar{\tau}$  be an upper bound such that:

$$\tau_t^k \leq \bar{\tau}, \quad \forall t \geq 0. \quad (15)$$

This bound translates into a constraint on the Euler equation for each group. Since a higher tax reduces the after-tax return, the maximum feasible tax implies a minimum possible gross return. For group 1 (capitalists), the Euler equation becomes:

$$u'(c_{1,t}) \geq \beta u'(c_{1,t+1}) [1 + (r_{t+1} - \delta)(1 - \bar{\tau})], \quad \forall t \geq 0. \quad (16)$$

The inequality reflects that the government cannot implement a tax rate higher than  $\bar{\tau}$ , so the after-tax return cannot fall below the level corresponding to  $\bar{\tau}$ .

**Intuition:** *If the unconstrained optimal policy would require  $\tau_t^k > \bar{\tau}$ , the bound becomes binding. The government then must accept a lower capital tax than desired from a Ramsey perspective, potentially shifting the burden to labour taxes or public debt.*

### 9.2. Consumption Floor

A second realistic constraint is that the government must ensure that no agent's consumption falls to zero (or below some minimum subsistence level). This is both a normative requirement (dignity) and a practical necessity (survival). We impose:

$$c_{1,t} \geq \underline{c}, \quad \forall t, \quad (17)$$

for some  $\underline{c} \geq 0$ . The same floor applies to group 2 indirectly because  $c_{2,t} = \lambda c_{1,t}$ , so the constraint automatically protects both groups if  $\lambda > 0$ .

**Intuition:** *Practically, this means the government cannot implement policies that lead to zero consumption for any agent. In the optimisation problem, this constraint will be enforced by a Lagrange multiplier  $\xi_t$  for each period.*

## 10. The Ramsey Problem

The government (Ramsey planner) chooses a Pareto optimal allocation. With two agents, a standard approach is to maximise the welfare of one group subject to a minimal utility level for the other group. This yields all points on the Pareto frontier.

### 10.1. Pareto Optimal Allocations

Any Pareto optimal allocation can be found by solving a problem where the planner maximises the utility of, say, consumer 1 (capitalists), subject to the constraint that consumer 2 (workers) achieves at least a certain utility level  $\underline{U}_2$ .

Thus, the planner solves:

$$\max_{\tau_0^k, \lambda, \{c_{1,t}, k_t, l_{1,t}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t (u(c_{1,t}) + v(l_{1,t}))$$

subject to:

$$\sum_{t=0}^{\infty} \beta^t (u(\lambda c_{1,t}) + v(\mathcal{K}(\lambda) l_{1,t})) \geq \underline{U}_2, \quad (13)$$

together with the feasibility constraint, the two implementability constraints (7) and (10), the tax limit (inequality Euler), and the consumption floor.

## 10.2. Lagrangian Formulation

Collecting all the above constraints, the Lagrangian for the planner's problem can be written as:

$$\begin{aligned} \mathcal{L} = & \sum_{t=0}^{\infty} \beta^t \left\{ u(c_{1,t}) + v(l_{1,t}) + \psi \left( u(\lambda c_{1,t}) + v(\mathcal{K}(\lambda) l_{1,t}) \right) \right. \\ & + \xi_t (c_{1,t} - \underline{c}) + \Delta_1 \left( u'(c_{1,t}) c_{1,t} + v'(l_{1,t}) l_{1,t} \right) \\ & + \Delta_2 \left( u'(c_{1,t}) \lambda c_{1,t} + \frac{\phi_2}{\phi_1} v'(l_{1,t}) \mathcal{K}(\lambda) l_{1,t} \right) \\ & + \gamma_t \left\{ u'(c_{1,t}) - \beta u'(c_{1,t+1}) [1 + (r_{t+1} - \delta)(1 - \bar{\tau})] \right\} \\ & + \mu_t \left[ F(k_{t-1}, e_t) + (1 - \delta)k_{t-1} - k_t - \frac{1 + \lambda}{2} c_{1,t} - g \right] \Big\} \\ & - \psi \underline{U}_2 - W, \end{aligned} \quad (18)$$

where  $W$  collects the initial-period terms from the implementability constraints:

$$W = u'(c_{1,0}) (\Delta_1 k_{1,-1} + \Delta_2 k_{2,-1}) [1 + (r_0 - \delta)(1 - \tau_0^k)]. \quad (20)$$

## 10.3. Interpretation of the Multipliers

- $\psi \geq 0$  is the Lagrange multiplier on the utility constraint of group 2. It measures how much the planner cares about workers relative to capitalists. The weight of group 1 is normalised to one.
- $\Delta_1$  and  $\Delta_2$  are the multipliers on the implementability constraints for groups 1 and 2, respectively. They capture the efficiency cost of distortionary taxation.
- $\xi_t \geq 0$  is the multiplier on the consumption floor. It is positive only if the floor binds.
- $\gamma_t \geq 0$  is the multiplier on the capital tax upper bound. It is positive only if the government would like to set  $\tau_t^k > \bar{\tau}$  but cannot.
- $\mu_t$  is the multiplier on the resource feasibility constraint (the shadow price of output).

## 10.4. Compact Notation for Optimality Conditions

From the first-order conditions of this Lagrangian, it is common to define two composite elasticities that summarise the effects of redistribution and distortion. Define:

$$\Omega^\ell \equiv 1 + \psi \mathcal{K}(\lambda)^{1+\sigma_l} + \left( \Delta_1 + \frac{\phi_2}{\phi_1} \mathcal{K}(\lambda) \Delta_2 \right) (1 + \sigma_l), \quad (21)$$



$$\Omega^c \equiv 1 + \psi \lambda^{1-\sigma_c} + (\Delta_1 + \lambda \Delta_2)(1 - \sigma_c). \quad (22)$$

**Intuition:** *These expressions appear in the modified Euler equations and labour supply conditions.  $\Omega^\ell$  adjusts the marginal disutility of labour for distributional concerns, while  $\Omega^c$  adjusts the marginal utility of consumption. The parameter  $\psi$  determines the planner's redistributive bias:*

- *If  $\psi$  is low, the planner favours capitalists (group 1).*
- *If  $\psi$  is high, the planner favours workers (group 2).*
- *As  $\psi \rightarrow \infty$ , the planner cares only about workers (Rawlsian extreme).*